

Behind the Curtains of LLMs: Boring Details, Major Consequences

Enrico Liscio



AlgoSoc /

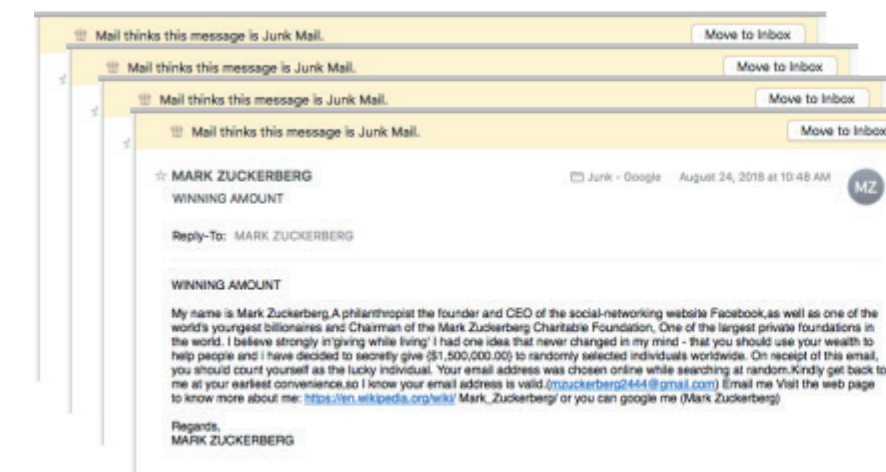
Machine Learning Refresher: Spam Detection

GOFAI

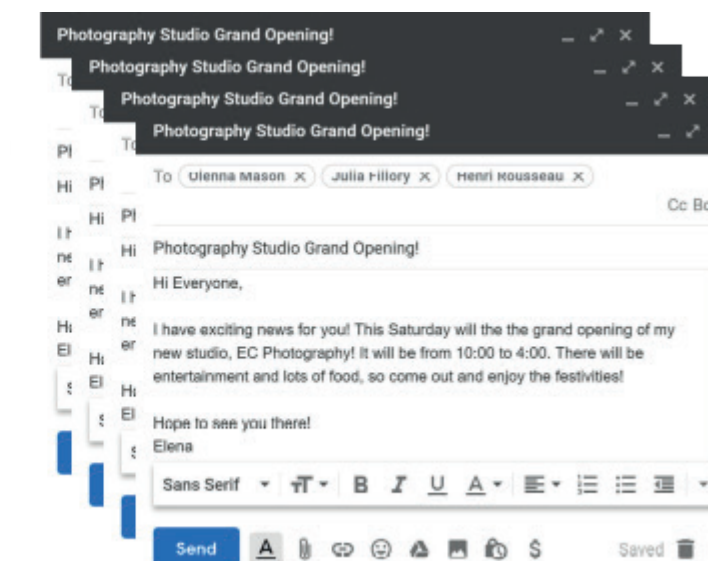
1. If the sentences end in multiple exclamation points -> spam;
2. If the sender is known -> not spam;
3. If a celebrity is mentioned -> spam;

...

Machine Learning



Spam



Not spam

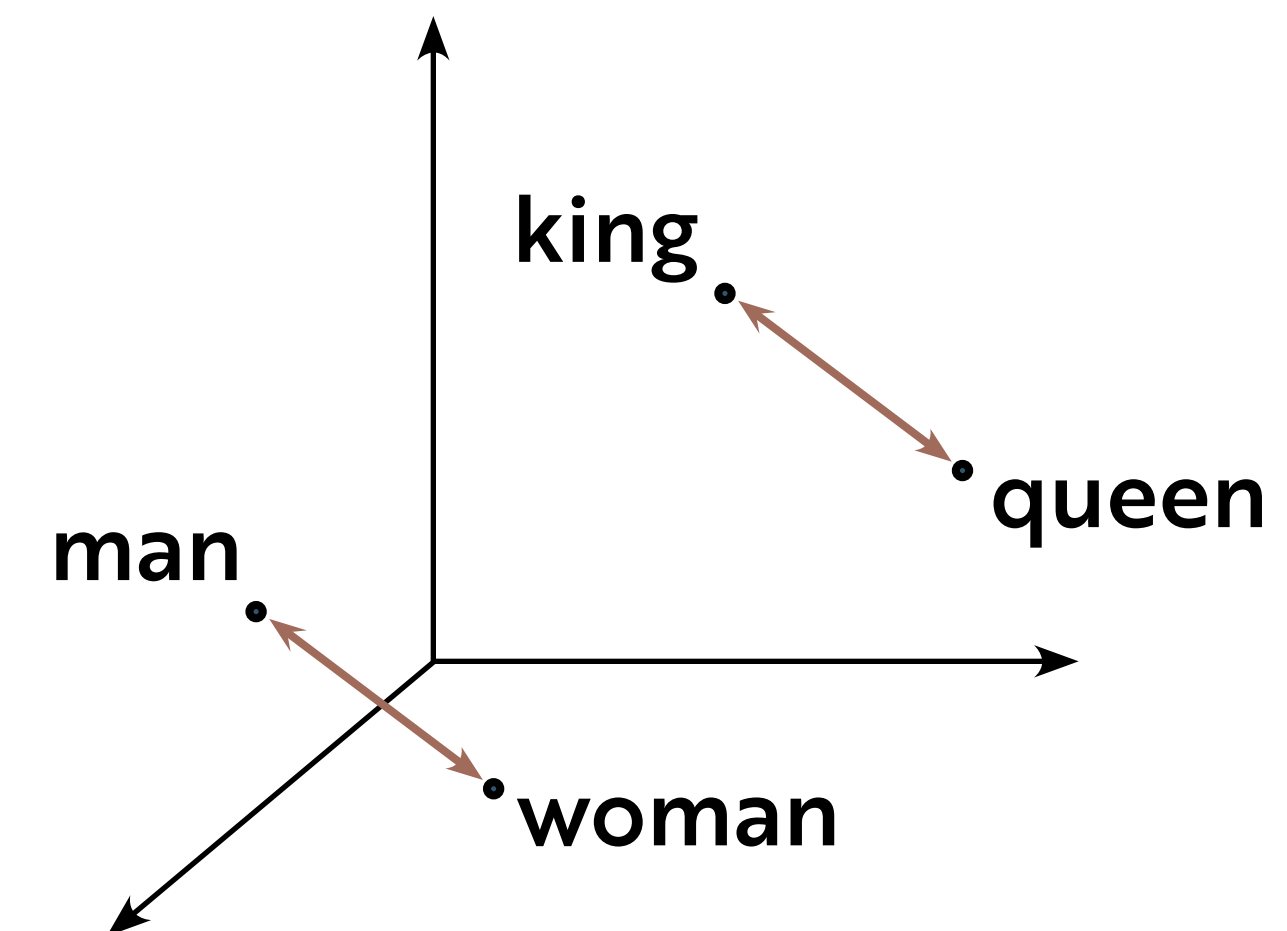
Word Embeddings

How to transform text in numbers that can be processed by algorithms?

Translate words into coordinates, aka *word embeddings*, popularized by word2vec (Google, 2013).

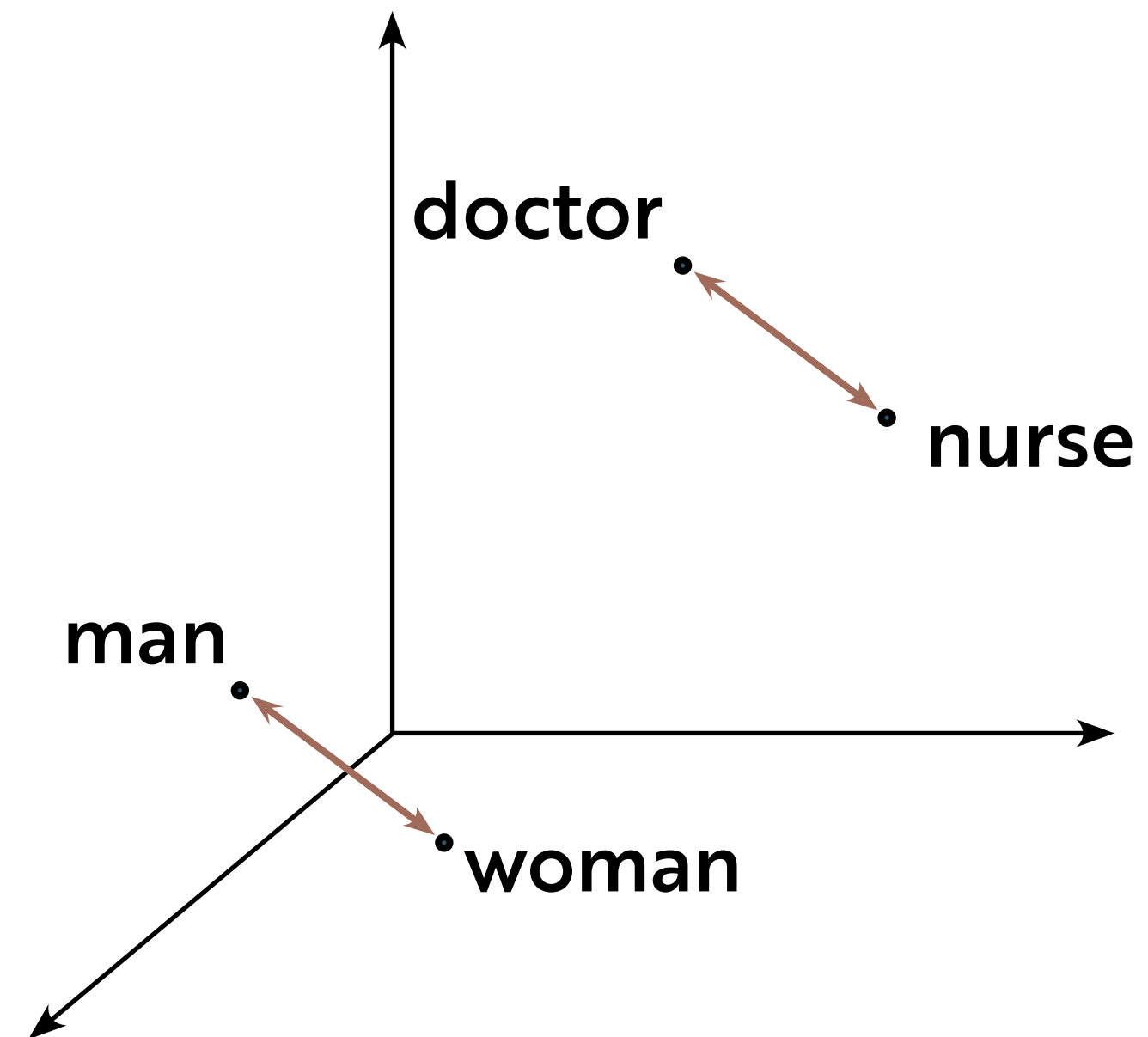
Words are mapped close together in space if:

- They co-occur in text;
- They appear in the same part of the sentence.



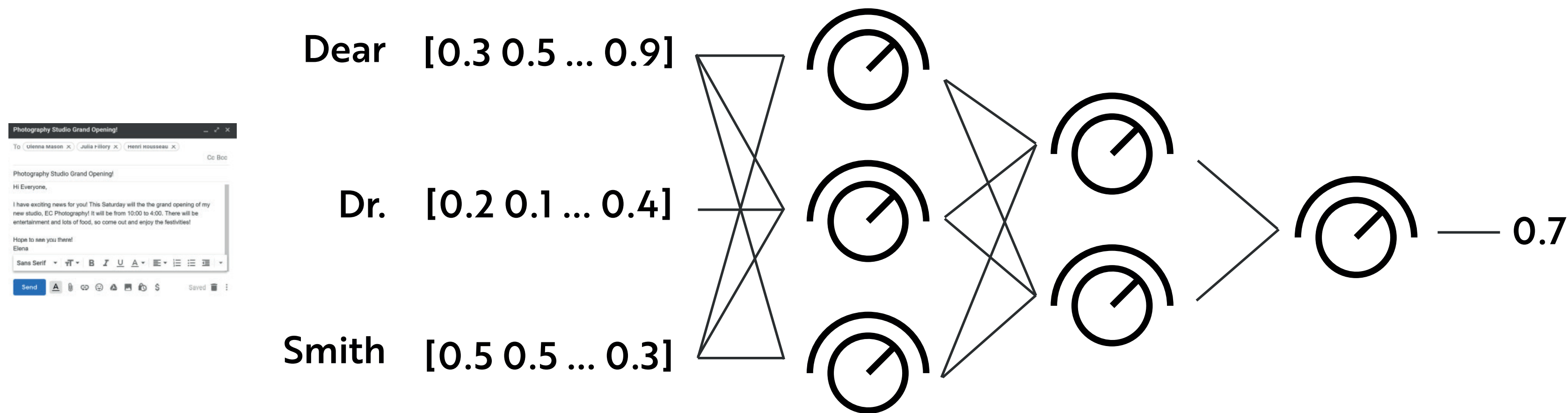
Word Embeddings

Word embeddings pick up all sorts of patterns, including **undersired biases**.



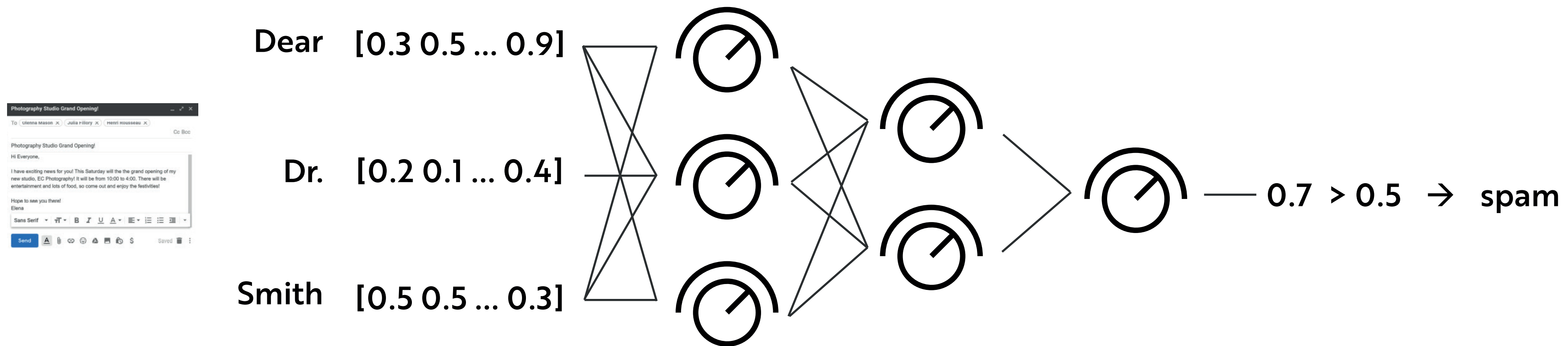
Transformers

Transformers (Google, 2017) are an architecture designed to process text (i.e., a sequence of word embedding vectors).



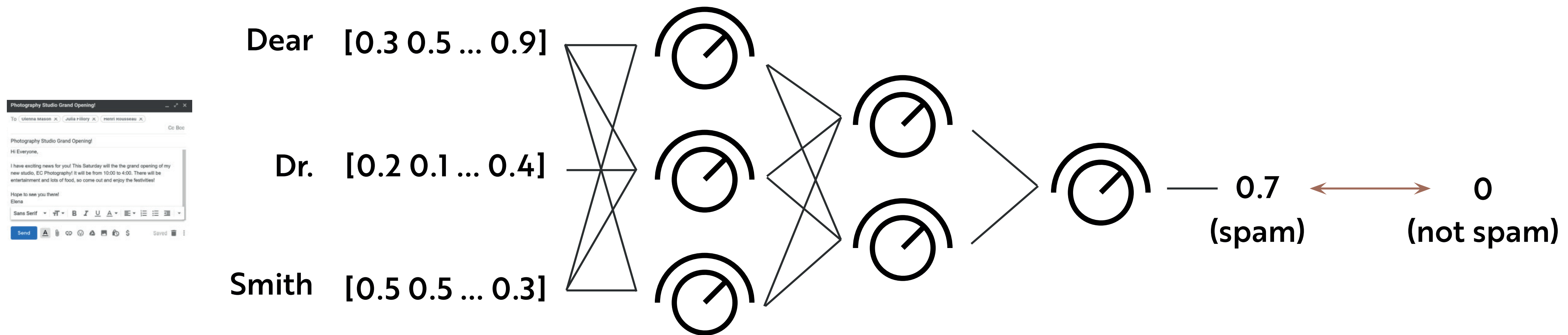
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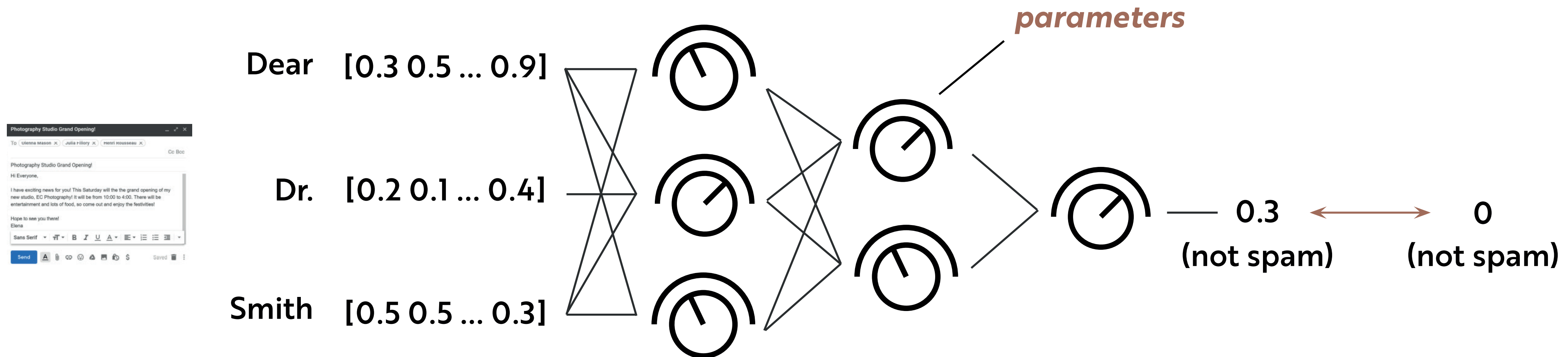
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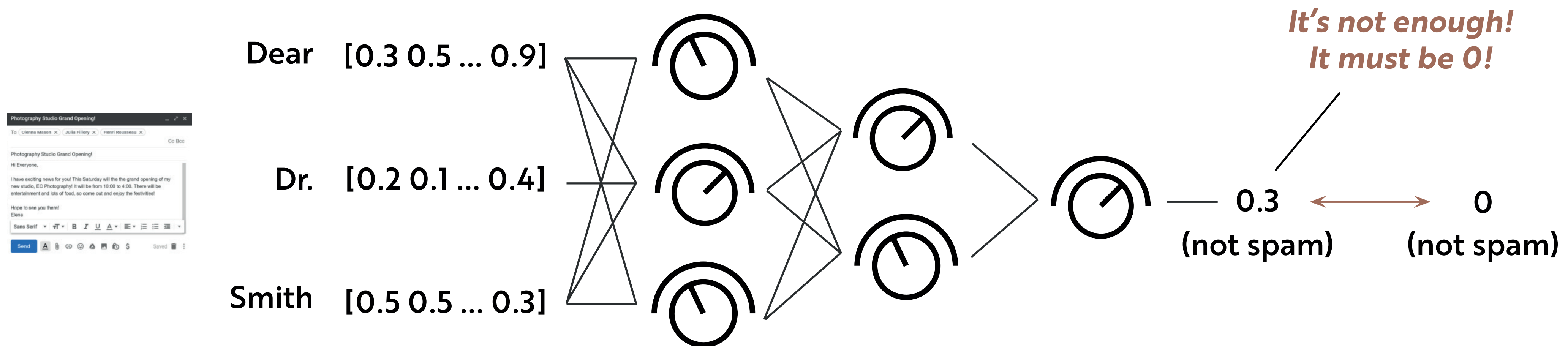
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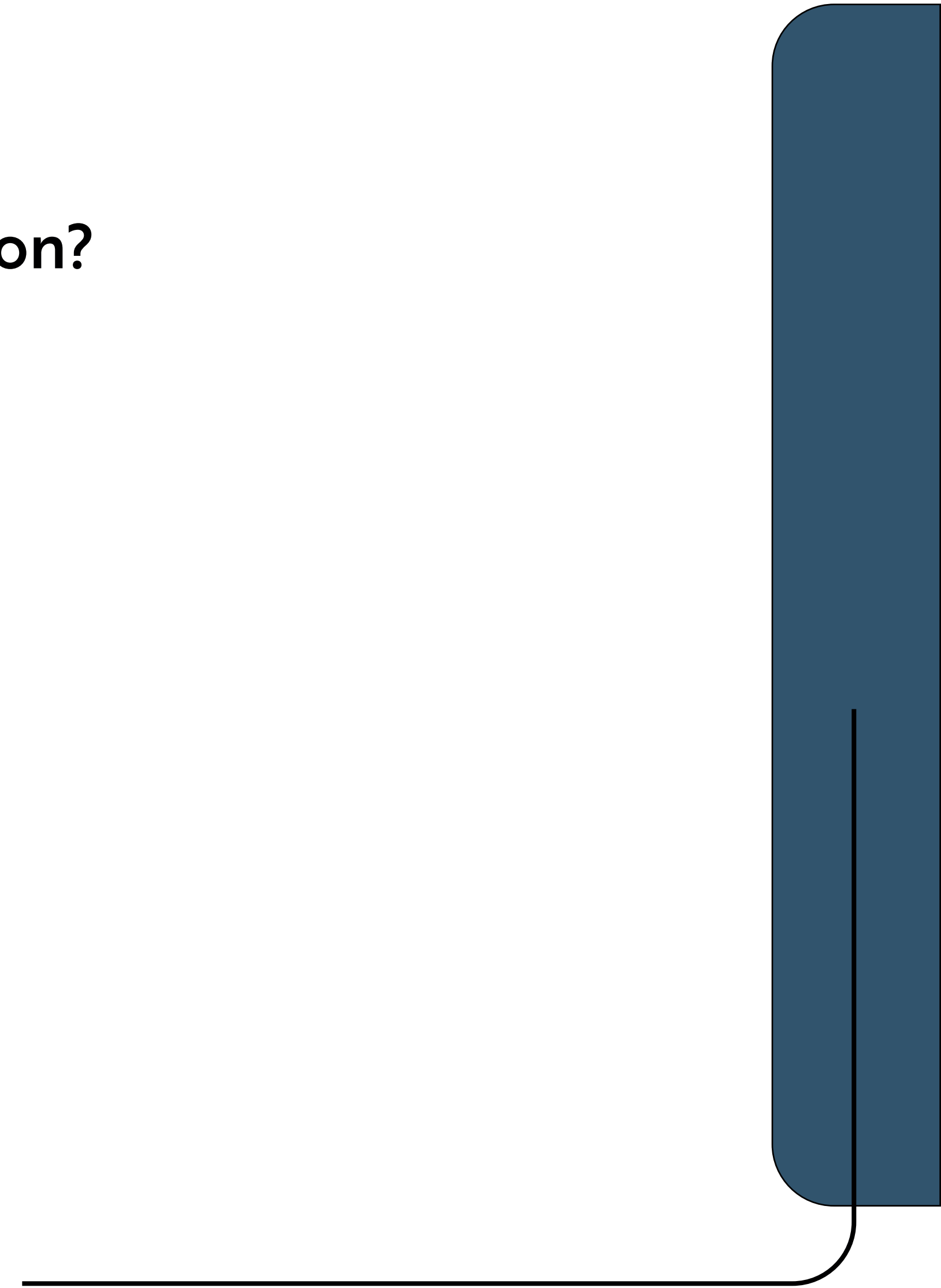
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Language Modeling

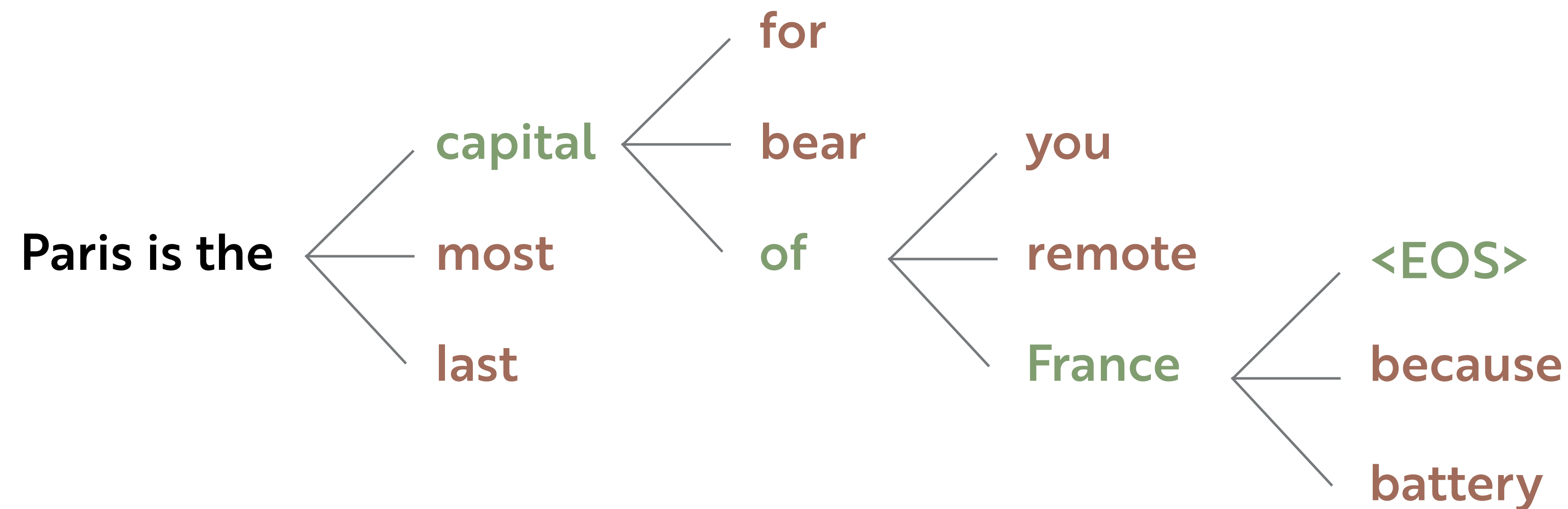
How to move beyond classification tasks to language generation?



Language Modeling

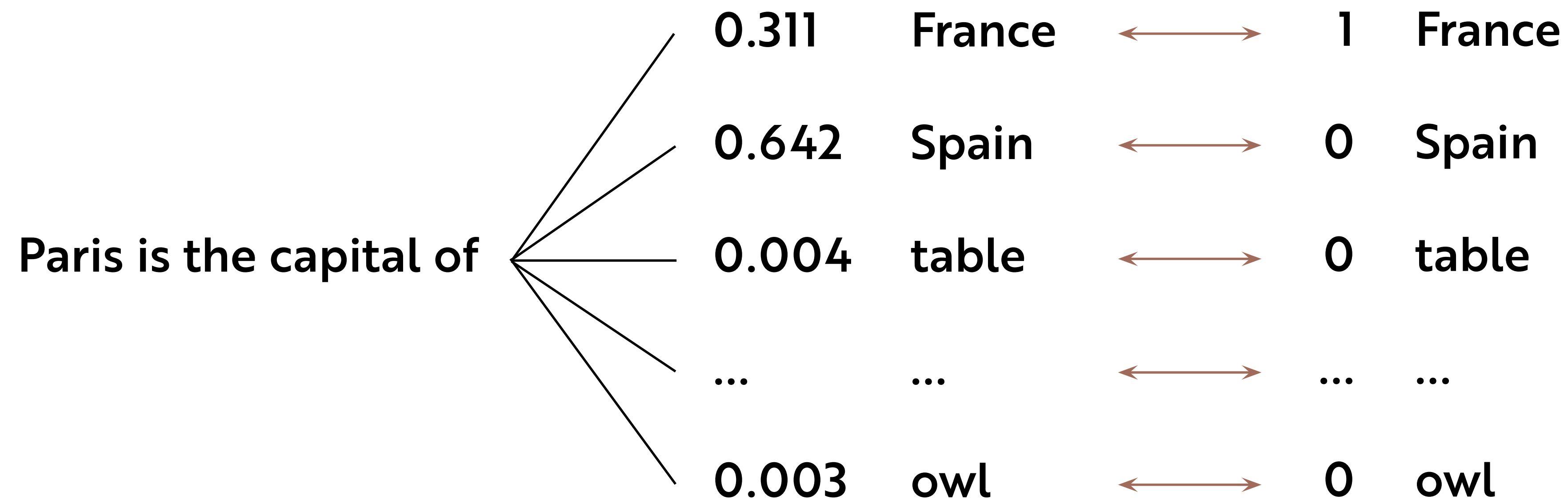
How to move beyond classification tasks to language generation?

GPT (OpenAI, 2018) shows that Transformers can be (pre-)trained with the **language modeling task**: predicting the next word in a sentence.



Language Modeling

The model predicts a **probability distribution** over all words in the dictionary (~50k words). This distribution is compared to the actual completion of the observed sentence.



Language Modeling

The same sentence is **observed with different conclusions** throughout the dataset, avoiding probability spikes to 0 or 1. However, **biased patterns are still picked up**.

Mary is a

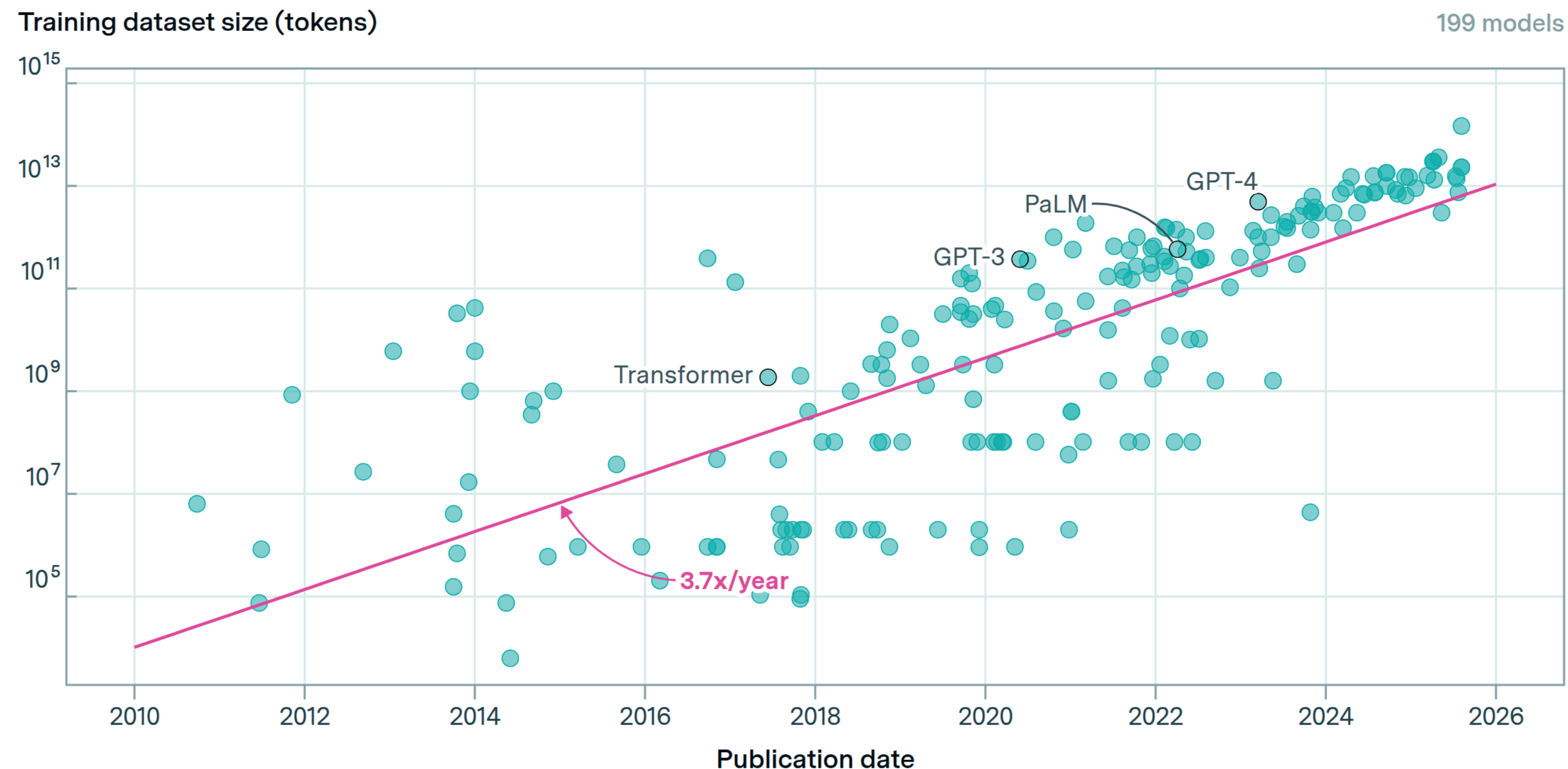
0.37	doctor	↔	1 doctor
0.41	nurse	↔	0 nurse
0.01	table	↔	0 table

Mary is a

0.45	doctor	↔	0 doctor
0.39	nurse	↔	1 nurse
0.01	table	↔	0 table

Language Modeling

Turns out that language modeling works very well at scale: **no explicit human labels are needed. Datasets and models explode in size.**



Robi Rahman and David Owen (2024), "The size of datasets used to train language models doubles approximately every six months". Published online at epoch.ai. Retrieved from: '<https://epoch.ai/data-insights/dataset-size-trend>' [online resource]

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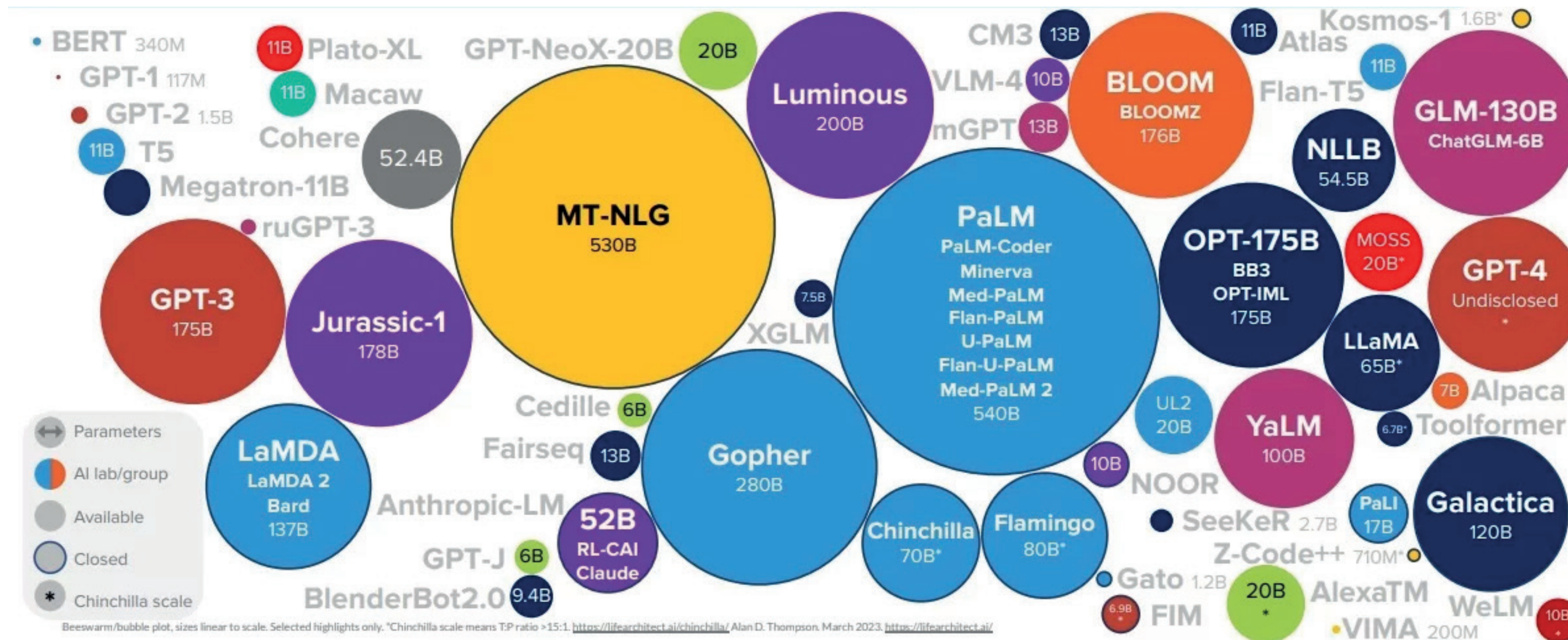


Image source: <https://pureinsights.com/blog/2023/what-are-large-language-models-llms-search-and-ai-perspectives/>

Language Modeling: Decoding

How to convert the probability distribution into a single word?

Intuitively, we would choose the word with the highest probability mass.

Paris is the capital of

- 0.87 France
- 0.06 Spain
- 0.01 table

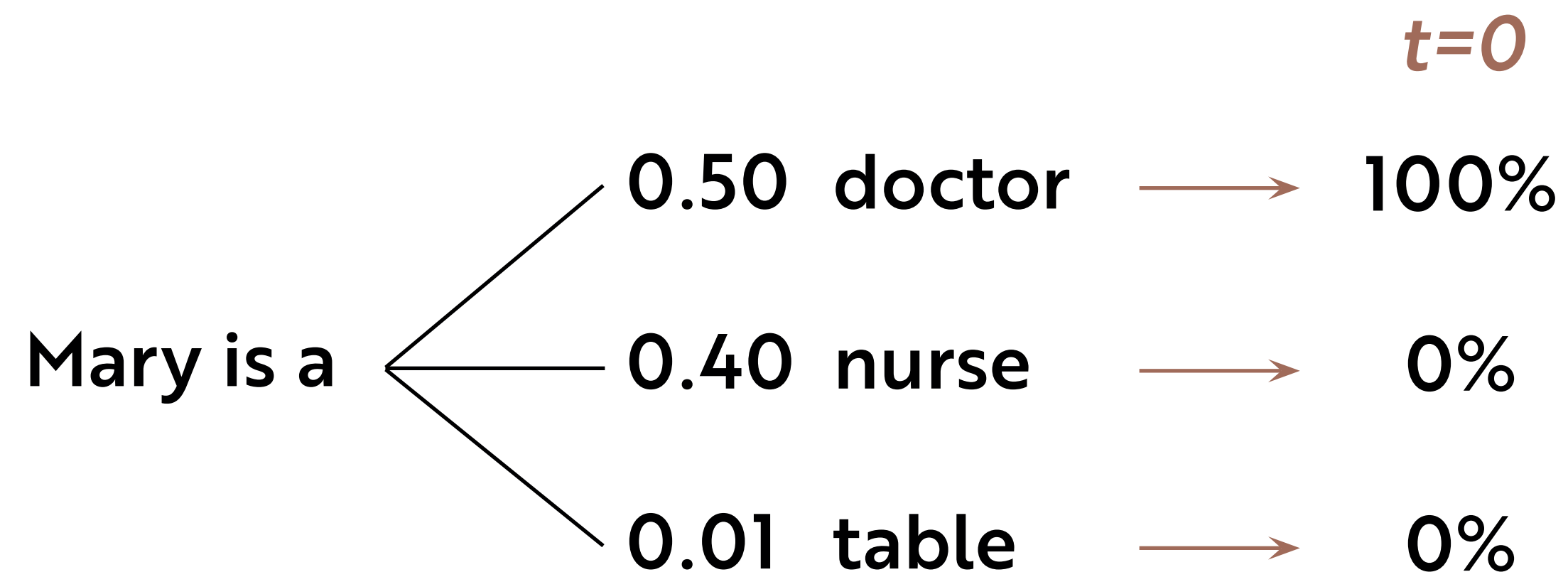
Mary is a

- 0.40 doctor
- 0.41 nurse
- 0.01 table

Language Modeling: Decoding

Choosing the word with highest probability is boring, repetitive, predictable.

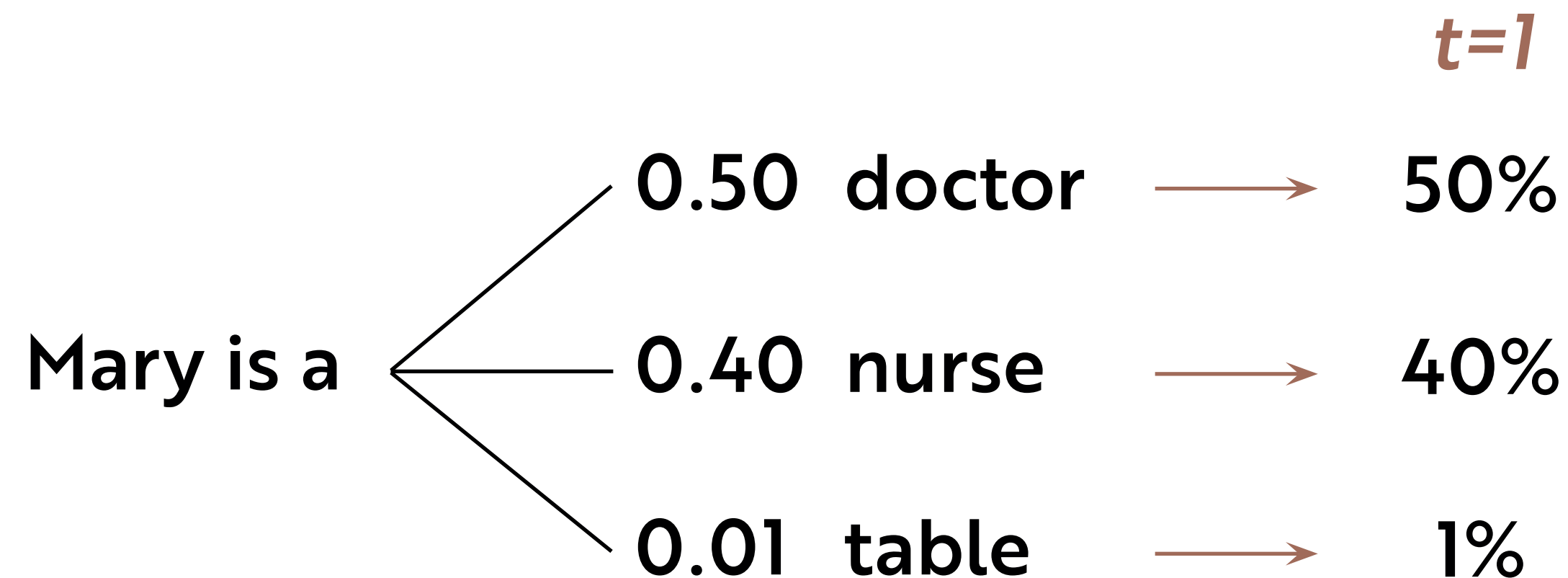
The *temperature* parameter is introduced to increase *creativity*.



Language Modeling: Decoding

Choosing the word with highest probability is boring, repetitive, predictable.

The *temperature* parameter is introduced to increase *creativity*.



Language Models are Not Chatbots

Pre-trained LLMs (and Transformers) are language models.
They **complete sentences** rather than answering questions.

"Can you summarize
this text?"



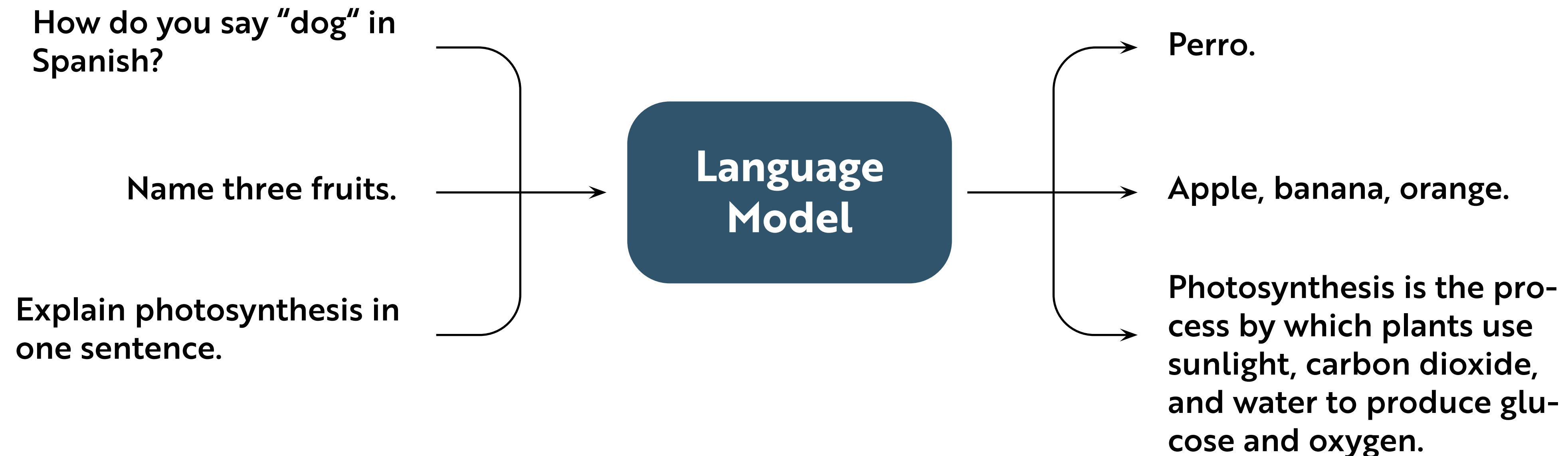
**Pre-trained
LLM**



"Write a summary
shorter than 200 words."

Instruction Tuning

Instruction tuning (Google, 2022) provides [question, answer] training examples to transform the model into an **instruction follower** (or, an assistant).



Alignment

Instruction tuning datasets contain dry Q&A content.
The alignment training phase aims at making models **kind, polite, friendly** (often framed as HHH, “helpful, honest, harmless”).

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Models are trained on **pairwise preferences** datasets, e.g., with RLHF (OpenAI, 2022).

“I failed an important exam and feel really stupid.”

A

“Failing an exam doesn’t define your intelligence. A lot of capable people struggle sometimes. If you want, we can talk through what went wrong or make a plan for next time.”

B

“Many students fail exams. Statistically, this happens often. You should review the material and prepare better next time.”

Christiano, Paul F., et al. “Deep reinforcement learning from human preferences.” NeurIPS, 2017.

Ouyang, Long, et al. “Training language models to follow instructions with human feedback.” NeurIPS, 2022.

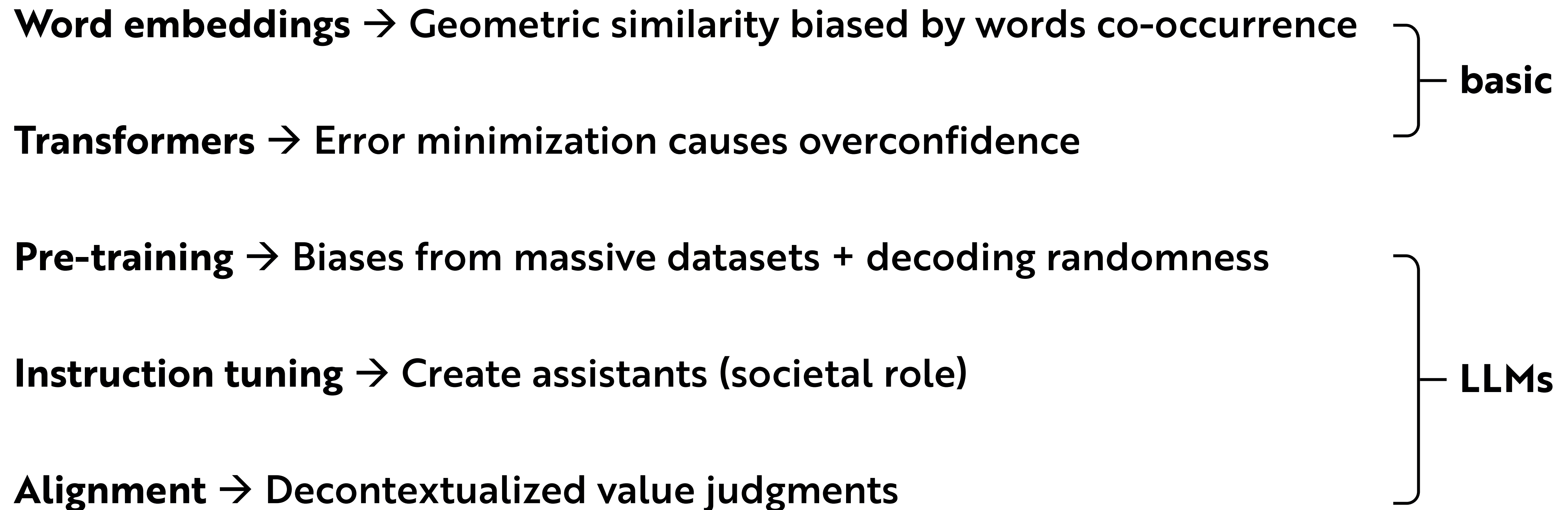
Recap

Word embeddings → Geometric similarity biased by words co-occurrence

Transformers → Error minimization causes overconfidence

basic

Recap

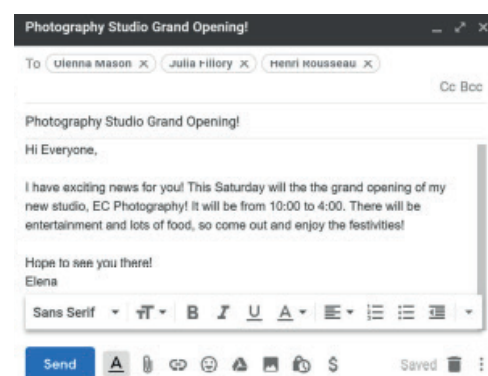


General Purpose Language Models

Through the three training steps, LLMs become **general purpose instruction followers**.

No need to specialize on a task of interest, just ask the model in a conversational fashion.

“Is this email spam?”



“This email may be considered spam because it contains promotional content and”

Public Values

Where do public values fall in the **safety vs. value alignment spectrum**?

Are public values derived from **rules**, or do they emerge from **observations**?

At **which training stage** can/should we instill public values?

Can we even instill public values during training (before deployment), or are public values **emergent in user-LLM interaction**?

Thank you!

Questions?

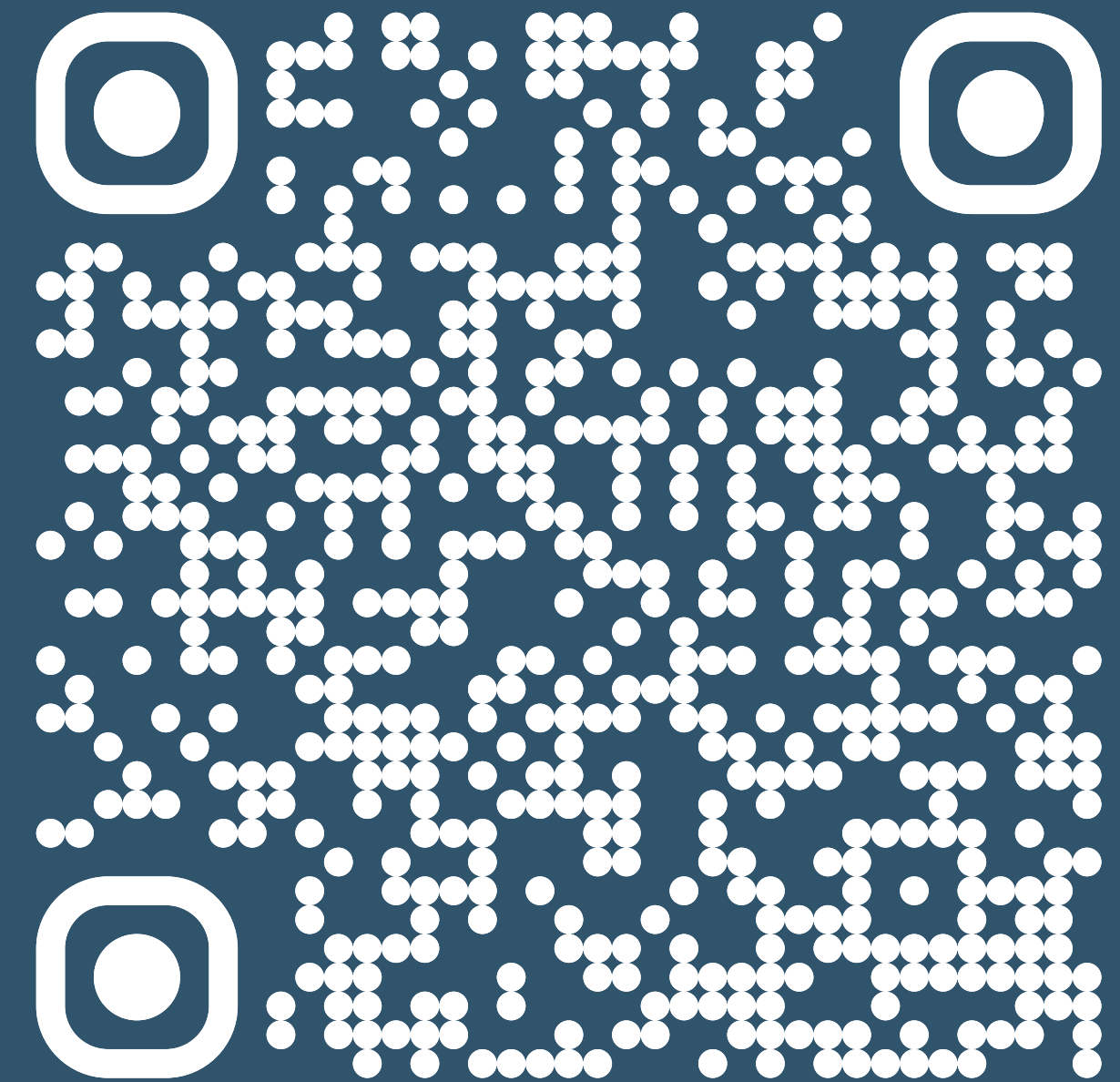


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Link to the presentation!



Language Modeling: Decoding

Top-p determines the probability cutoff for words to be sampled from.

